24CS31 — Last-Minute Prediction Sheet

Laplace Transforms & Vector Space · SEE 100 marks / 3 hrs · **Answer one full question from each unit** · Q1|Q2 → Unit I Q3|Q4 → II Q5|Q6 → III Q7|Q8 → IV Q9|Q10 → V

THE 60-SECOND RULE. Don't read all ten questions. In each unit, find the **one trigger** below and answer **that** question — the rest of its parts are the ones you studied. **I:** the t^n proof • **II:** the Heaviside piecewise • **III:** the reflection→rotation→contraction question • **IV:** QR + least-squares (that's Q8) • **V:** orthogonally diagonalize.

READ THE TWO RIGHT-HAND COLUMNS SEPARATELY — they mean different things.
"Topic came up" = how many of the four papers asked *this kind of question at all*, with any numbers. **"SAME values"** = how many asked it with the *identical matrix, function or constants* printed here. A row reading **4/4 topic / 0/4 same** (e.g. QR factorization) is **certain to appear but the matrix will be new** — you must own the method, not an answer. A row reading **4/4 / 3/4** (orthogonal diagonalization) is a question you can rehearse end-to-end and reproduce.
Colour is just the size of the number: 4/4 3/4 2/4 1/4 0/4. Proofs count as "same values" when the wording is identical. The **2-mark definition slot only exists in the March 2024 format**, so it can never score above 1/4 — it is still near-certain under the current pattern.

UNIT I — Laplace Transforms

20 marks · a(2) + b(4) + c(7) + d(7)

PICK the question containing the t^n proof. The sin $\sqrt{t}$ question is **always** in the same one (4/4 papers) — that's 11 marks together. The periodic-function problem is always in the *other* one.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	**Prove** $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\{F(s)\}$, n a positive integer. **← THE trigger. Do this first.**	07	Q1c or Q2b	4/4	4/4
2	Given $L\{\sin\sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, **show** $L\{\frac{\cos\sqrt{t}}{\sqrt{t}}\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$	04	Q1b — with #1	4/4	2/4
3	**Evaluate** (i) $L\{t\, e^{-4t} \sin 3t\}$ (ii) $L\{\frac{t - \sinh at}{t}\}$ (multiply by $t \to -F'(s)$; divide by $t \to \int_s^\infty F$)	07	Q1d or Q2c	4/4	1/4
4	**Evaluate** (i) $\int_0^\infty \frac{e^{-t}\sin t}{t} dt$ (*ans* $\pi/4$) (ii) $L\{t \sin 3t \cos 2t\}$	07	Q1d	3/4	1/4
5	**Periodic function.** Triangular wave, period $2a$: $f(t) = \begin{cases} t, & 0 \le t \le a \\ 2a - t, & a \le t \le 2a \end{cases}$ *ans* $\frac{1}{s^2}\tanh(\frac{as}{2})$ OR half-rectifier $E\sin(\omega t)$ on $(0,\pi/\omega)$, 0 after	07	OTHER question	4/4	2/4
6	**Obtain** $L\{(\sqrt{t} + 1/\sqrt{t})^3\}$ (expand first, then term-by-term)	04	Q2b	2/4	1/4
7	**Definition (2 marks — free).** "Write the Laplace transform of a periodic function" = $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t)\,dt$ • or "Define the Laplace transform"	02	Q1a & Q2a	2/4	1/4

UNIT II — Application of Laplace Transforms

20 marks · a(2) + b(4) + c(7) + d(7)

PICK the question containing the Heaviside piecewise function. An ODE problem is **always** sitting next to it (4/4 papers) — that's **14 guaranteed marks** in one question. Convolution is in the other one.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	**Express in terms of the Heaviside function and find its LT:** $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ **← THE trigger. Same values twice.**	07	Q3c	4/4	2/4
2	**Solve by Laplace transforms** (one of these, in the *same* question as #1): • $dx/dt - 2y = \cos 2t$; $dy/dt + 2x = \sin 2t$, $x(0)=1$, $y(0)=0$ • $y'' + 4y' + 3y = e^{-t}$ (set twice, different ICs) • LR circuit: $L\, di/dt + Ri = Ee^{-at} \to i = \frac{E}{R-aL}(e^{-at} - e^{-Rt/L})$	07	Q3d / Q4d — with #1	4/4	2/4
3	**Convolution theorem.** $L^{-1}\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\}$ OR verify it for $f_1=t$, $f_2=\cos t$	07	OTHER question	4/4	2/4

4	**Evaluate** $L^{-1}\{\log\frac{s^2+1}{s(s+1)}\}$ differentiate F, then $L^{-1}\{F\} = -t\,f(t)$	04	Q3b	3/4	2/4
5	**Find** $L^{-1}\{\frac{3s+2}{s^2-s-2}\}$ — partial fractions. Every paper carries a short inverse LT in *both* options.	04	Q4b	4/4	1/4
6	**Definition (2 marks — free).** Define the unit step function + graph • Define the Dirac-delta function + sketch	02	Q3a & Q4a	1/4	1/4

UNIT III — Vector Space and Linear Transformation

20 marks · a(6) + b(7) + c(7)

PICK the question containing the reflection→rotation→contraction problem (usually **Q6**). In the two newest papers that question also held the transition matrix *and* kernel&range — three 4/4 topics, a clean 20.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	**Matrix of a composition + image of a point.** Reflection in the x-axis, then rotation π/2, then contraction of factor 1/3. Find the image of $\begin{bmatrix}4\\1\end{bmatrix}$. **← THE trigger.** Multiply **right to left**: $k\,R_\theta\,Ref$. Reflect in x = diag(1,−1); in y = diag(−1,1).	06	Q6a	4/4	1/4
2	**Transition matrix.** $B = \{(1, 2), (3, -1)\}$, $B' = \{(3, 1), (5, 2)\}$ of R^2. Find the transition matrix from B to B'; if $[u]_B = \begin{bmatrix}2\\1\end{bmatrix}$, find $[u]_{B'}$. **Same bases twice running.**	07	Q6b — with #1	4/4	2/4
3	**Kernel and range + verify rank-nullity** for $A = \begin{bmatrix}1&2&3\\0&-1&1\\1&1&4\end{bmatrix}$ (rank 2, nullity 1). Be ready for the formula version too, e.g. $T(x,y) = (3x, x-y, y)$.	07	Q6c — with #1, #2	4/4	1/4
4	**Prove T is linear + find images.** $T{:}P_2\to P_2$, $T(ax^2+bx+c) = (a+b)x^2+c$; image of $5x^2+6x+1$. OR $T(x,y) = (3x+y, 2y, x-y)$; images of (1,2), (2,−5)	07	Q5b — all 4 papers	4/4	2/4
5	**Span / linear combination / independence / basis.** e.g. is a set of four 2×2 matrices a basis of M_{22}? (flatten each to a 4-vector, take the 4×4 determinant) • or: does {(1,2,3), (−1,−1,0), (2,5,4)} span R^3?	06	Q5a	4/4	1/4
6	**Prove** $T(x) = Ax$ rotates x through θ about the origin if $A = \begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$ — bookwork escape hatch if Q6 looks ugly.	07	Q5c	2/4	2/4

UNIT IV — Orthogonal Projections

20 marks · **Q7 = 6+7+7 \| Q8 = 10+10**

Q8 is QR + least-squares — both 4/4, and that's the whole 20. Take it if your algebra holds up. **Q7 is the safer three-problem question** — and the four-subspaces question is *always* in Q7 (4/4).

#	Question — write this	Marks	Section	Topic came up	SAME values
1	**QR factorization.** Mar 24 used $\begin{bmatrix}3&-5&1\\1&1&1\\-1&5&-2\\3&-7&8\end{bmatrix}$. Matrix changes every paper — own the method: Gram–Schmidt the columns, normalise → Q, then $\boldsymbol{R = Q^TA}$.	10	Q8a	4/4	0/4
2	**Least-squares solution of $AX = b$:** $A = \begin{bmatrix}1&3&5\\1&1&0\\1&1&2\\1&3&3\end{bmatrix}$ $b = \begin{bmatrix}3\\5\\7\\-3\end{bmatrix}$ **← identical values, twice.** Solve $A^TAx = A^Tb$.	10	Q8b — with #1	4/4	2/4
3	**Dimension and basis for the four fundamental subspaces** of $A = \begin{bmatrix}1&2&0&1\\0&1&1&0\\1&2&0&1\end{bmatrix}$ **← same matrix in 3 of 4 papers.** row 3 = row 1 ⇒ rank 2; dim nul = 2, dim left-nul = 1	07	Q7 — always	4/4	3/4
4	**Complete solution of $Ax = b$.** Mar 24: $x+3y+3z=1$, $2x+6y+9z=5$, $-x-3y+3z=5$. Answer as $x = x_p + c_1n_1 + \ldots$	06	Q7a	3/4	0/4
5	**Orthogonal projection.** $y = \begin{bmatrix}2\\3\end{bmatrix}$, $u = \begin{bmatrix}4\\-7\end{bmatrix}$. Find proj of y onto u; split y into a part in span{u} and a part orthogonal to it. proj = $((y\cdot u)/(u\cdot u))u$ — 5 minutes, 7 marks	07	Q7b	1/4	1/4
6	**Gram-Schmidt orthonormal basis** — e.g. span{(1,1,1,1), (0,1,1,1), (0,0,1,1)}. You already know this from QR: just stop before forming R.	10	Q8a substitute	1/4	1/4

UNIT V — Applications of Eigenvalue Decomposition

20 marks · a(5) + b(5) + c(10)

PICK the question containing "orthogonally diagonalize" (Q10 in Mar 24). The two property-checks travel with it in 3 of 4 papers — 10 + 5 + 5, and the 5-markers are ten minutes total. SVD sits in the *other* question.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	**Orthogonally diagonalize** $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ **← THE trigger. Same matrix, 3 of 4 papers.** eigenvalues **2, 3, 6** — all distinct, so eigenvectors come out orthogonal on their own. Normalise → P, then $P^T AP = D$.	10	Q10c	4/4	3/4
2	**Is** $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ **positive definite?** minors 1, 0, 0 → **positive semi-definite, NOT positive definite**. Say so explicitly.	05	Q10a — with #1	4/4	2/4
3	**Is** $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ **unitary?** Show $A^* A = I$ (**yes**) OR the Hermitian version: is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian? (**yes**)	05	Q10b — with #1, #2	4/4	2/4
4	**Find the SVD of** $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$ $A^T A = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$, eigenvalues 18 and 2 ⇒ singular values $3\sqrt{2}$ and $\sqrt{2}$	10	OTHER question (Q9c)	4/4	2/4
5	**Quadratic form → no cross-product terms.** $Q(x) = x_1{}^2 - 8x_1x_2 - 5x_2{}^2$. Build the symmetric matrix, orthogonally diagonalise, substitute $x = Py$.	05	Q9a	2/4	0/4
6	**Steady state of the Markov matrix** $A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$ solve $(A-I)x = 0$, scale so entries sum to 1	05	Q9b	1/4	1/4
7	**Diagonalize and hence find A^n** ($A^n = PD^nP^{-1}$). Appeared 3 papers straight, then **skipped in Mar 24** — which makes it **overdue**, not safe.	10	Q9 (10-mark slot)	3/4	0/4

IF YOU HAVE 20 SPARE MINUTES — the one hole in your prep. PCA is named in your syllabus *and* in CO5, and has **never** been asked in four papers. If it comes, it's 5 marks in Unit V, most likely next to the SVD. All you need: centre the data (subtract the column means), form the covariance matrix $C = \frac{1}{n-1} X^T X$, find its eigenvalues and eigenvectors, take the eigenvector of the largest eigenvalue as the first principal component, and report $\lambda_1/\Sigma\lambda_i$ as the variance it explains. **Conic sections** is the same gap — diagonalise the quadratic form; both eigenvalues the same sign → ellipse, opposite signs → hyperbola.

How to read this: "Section" is the slot the question occupied in the March 2024 paper (the only one under your current course title) with the older papers cross-checked. "Freq" is how many of the four papers carried that topic. Items marked **identical values** have been set with the same numbers more than once. Predictions are inferences from four papers — strong, not guaranteed. Full 22-page analysis, pairing laws and the complete question index are in *24CS31_Prediction_Sheet.pdf.*